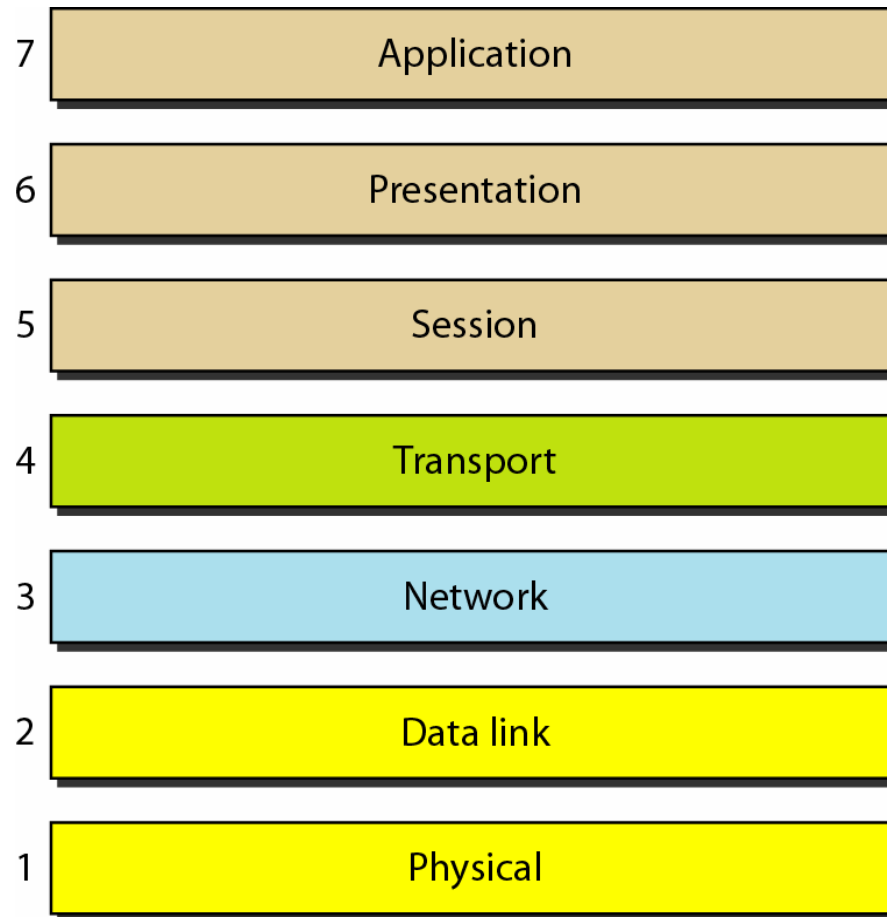


Midterm exam 061204

Solutions

1. Name all the layers in the OSI-model!



**2. Show the shortest form of the IPv6 address
0340:0000:0000:0000:000B:C003:0234**

340::B:C003:234

3. Name two security protocols used in Internet.

IPSec, SSL/TLS, PGP, SSH

4. What is the function of DHCP?

The Dynamic Host Configuration Protocol (DHCP) provides static and dynamic IP address allocation.

That is: DHCP gives a computer an IP address when it's connected to a LAN.

5. What is message integrity (data integrity) and how can it be provided?

Message Integrity means that the message must arrive at the receiver exactly as it was sent. There must be no changes during the transmission, neither accidentally nor maliciously.

It can be provided by a message digest that is a special file generated from the message. If the message is changed, the sender will not be able to reproduce the message digest and can thereby detect that someone tampered with the message.

6. Which protocol is used by ping?

ICMP (Internet Control Message Protocol)

7. Name two problems that IPv4 suffers from.

- Lack of addresses.
- No support for real-time applications.
- No security functions.
- Variable header length (makes routing more capacity consuming).
- No feasibility to extend the protocol.

8. On which protocol layer does a repeater work on?

The physical layer.

It amplifies signals coming on one link element and transmits them on the next link element.

9. Give two examples of link metrics (path costs)

Hop count, inverse of bandwidth, delay.

10. Which of the routing techniques Distance Vector and Link State converges the fastest?

Link State

Link State sends Link State Advertisements to **all** nodes/routers, which means that the convergence is more or less instant when using Link State.

The hop-by-hop information flow used in Distance Vector slows down the information spreading function considerably, and therefore convergence time is much longer when using Distance Vector.